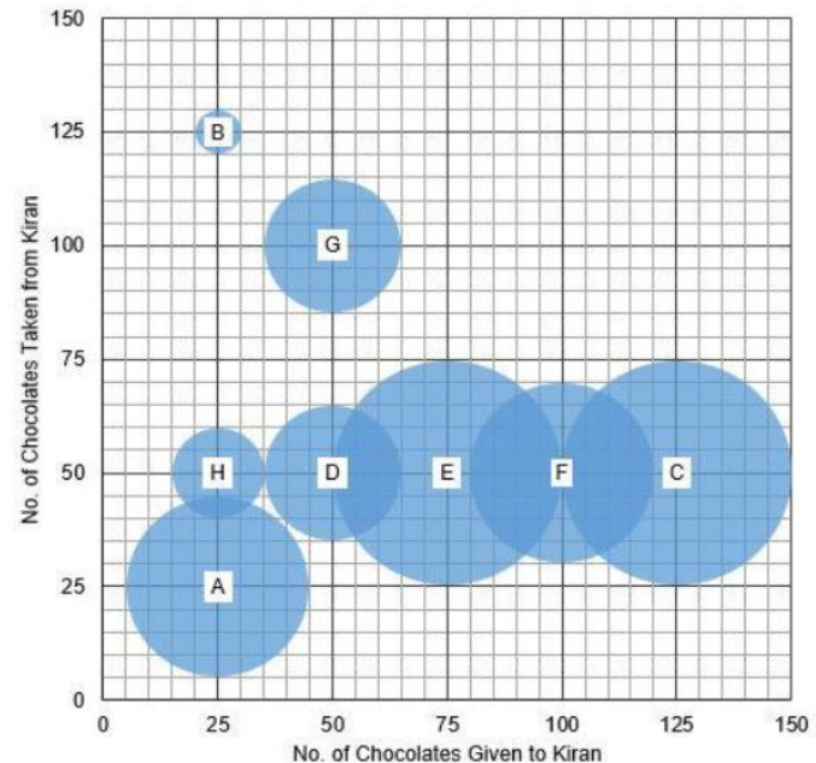


DIRECTIONS for questions 1 to 5: Answer these questions on the basis of the information given below.

Kiran, a budding data analyst, had exactly 50 chocolates with him at the beginning of a day. During the day, eight children, A through H, visited him one after the other, not necessarily in the same order. Each child first took some chocolates that Kiran already had, then gave him some chocolates and immediately left. Also, the number of chocolates taken or given by each child was an integer. Further, C was not the first child to visit him.

At the end of the day, Kiran made the following bubble chart in which each bubble represents the number of chocolates with him after each child left. The horizontal axis represents the number of chocolates given to him and the vertical axis represents the number of chocolates taken from him. The width (diameter) of each bubble represents the number of chocolates left with him after each child (indicated by the label on the bubble) left..



Q1. DIRECTIONS for question 1: Type in your answer in the input box provided below the question.

How many chocolates did Kiran have after the sixth child left?

Q2. DIRECTIONS *for question 2:* Select the correct alternative from the given choices.

Who was the fifth child to visit Kiran?

☐ a) D

☐ b) G

☐ c) E

☐ d) H

Q3. DIRECTIONS *for question 3:* Type in your answer in the input box provided below the question.

How many chocolates did Kiran have immediately after the first child left?

Q4. DIRECTIONS *for questions 4 and 5:* Select the correct alternative from the given choices.

How many children visited Kiran before H did?

- ☐ a) 4
- ☐ b) 6
- ☐ c) 5
- ☐ d) None of the above

Q5. DIRECTIONS *for questions 4 and 5:* Select the correct alternative from the given choices.

Among the children who visited before C, how many children gave Kiran more than 50 chocolates?

☐ a) 3

☐ b) 1

☐ c) 0

☐ d) 2

DIRECTIONS *for questions 6 to 10:* Answer these questions on the basis of the information given below.

In a colony, there are exactly three buildings – B1, B2 and B3. The number of persons living in each of the three buildings is the same and they have exactly four types of vehicles – Motorcycles, Sedans, SUVs and Hatchbacks. Any person who has an SUV can also have a Motorcycle or a Hatchback but not both. Everyone else owns exactly one type of vehicle. Further, no one owns more than one vehicle of any type.

It is also known that

i. the number of persons who own two vehicles in B1, B2 and B3 are in the ratio 2:3:5, while the number of persons who own a motorcycle in B1, B2 and B3 are in the ratio 6:7:8.

ii. the total number of persons who own a Sedan across the three buildings is the same as the number of persons who own only an SUV in B2.

iii. the total number of persons living in each building is a multiple of 10 and no one owns a Sedan in B3.

iv. the total number of persons who own only Motorcycles across the three buildings is a multiple of 15 and is 15 more than the number of persons who own a hatchback in B3.

v. 40 persons own only Hatchbacks in B3, which, in turn, is twice the total number of persons who own an SUV across the three buildings.

vi. exactly nine persons own only an SUV in B2, while the number of persons who own only Motorcycles in B2 is 20.

Q6. DIRECTIONS *for question 6:* Type in your answer in the input box provided below the question.

How many persons own only a Motorcycle in B1?

Q7. DIRECTIONS *for questions 7 to 9:* Select the correct alternative from the given choices.

What BEST can be said about the total number of vehicles in B2?

- ☐ a) It lies between 70 and 79.
- ☐ b) It is exactly 70.
- ☐ c) It lies between 67 and 71.
- ☐ d) It is exactly 73.

Q8. DIRECTIONS *for questions 7 to 9:* Select the correct alternative from the given choices.
What is the total number of hatchbacks owned by all the persons across the three buildings?

- ☐ a) 115
- ☐ b) 121
- ☐ c) 124
- ☐ d) 128

Q9. DIRECTIONS *for questions 7 to 9:* Select the correct alternative from the given choices.

What BEST can be said about the number of SUVs owned by all the persons in B3?

- ☐ a) It is either 5 or 6.
- ☐ b) It is exactly 5.
- ☐ c) It is exactly 6.
- ☐ d) It is either 6 or 7.

Q10. DIRECTIONS *for question 10:* Type in your answer in the input box provided below the question.

If the sum of the number of persons who own a sedan in B1 and the number of persons who own only a hatchback in B2 is the same as the sum of the number of persons who own only a hatchback in B1 and the number of persons who own a sedan in B2, how many persons own a hatchback in B2?

DIRECTIONS for questions 11 to 15: Answer these questions on the basis of the information given below.

In a college, five professors, Agarwal, Banerjee, Das, D'Souza and Pandey, were part of a panel to select the Student of the Year. Five students, Shankar, Tarun, Utsav, Varun and Wasim, were nominated for this award. Each professor awards the five students a distinct number of points from 1 to 5 (both inclusive).

The following table provides the total number of points awarded by the five professors combined to the five students:

Student	Shankar	Tarun	Utsav	Varun	Wasim
Points	21	20	5	14	15

Also, Das awarded 2 points to Tarun, D'Souza awarded 5 points to Varun and Banerjee awarded 5 points to Shankar.

Q11. DIRECTIONS for questions 11 to 15: Select the correct alternative from the given choices.

How many points did Banerjee award Tarun?

- ☐ a) 5
- ☐ b) 4
- ☐ c) 3
- ☐ d) Cannot be determined

Q12. DIRECTIONS *for questions 11 to 15:* Select the correct alternative from the given choices.
To which of the following students did Banerjee and D'Souza award the same number of points?

- ☐ a) Shankar
- ☐ b) Tarun
- ☐ c) Varun
- ☐ d) Wasim

Q13. DIRECTIONS *for questions 11 to 15:* Select the correct alternative from the given choices.

How many professors awarded less than 4 points to Wasim?

☐ a) 1

☐ b) 2

☐ c) 5

☐ d) 4

Q14. DIRECTIONS *for questions 11 to 15:* Select the correct alternative from the given choices.
Which of the following students was not awarded exactly 2 points by any of the five professors?

- ☐ a) Shankar
- ☐ b) Wasim
- ☐ c) Varun
- ☐ d) Tarun

Q15. DIRECTIONS *for questions 11 to 15:* Select the correct alternative from the given choices.

It was found that, for one of the five professors, Professor X, the points that he awarded to Shankar should have been awarded to Tarun and vice versa. After the points were adjusted appropriately, Tarun had the highest number of points.

Who among the following could have been Professor X?

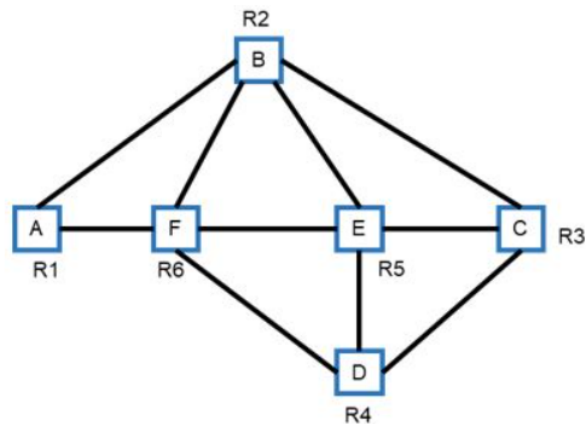
- ☐ a) Agarwal
- ☐ b) Pandey
- ☐ c) Das
- ☐ d) D'Souza

DIRECTIONS for questions 16 to 20: Answer these questions on the basis of the information given below.

In a mansion, there are six rooms – R1 through R6 – connected by different corridors. Each room has multiple doors each of which connect the room to each corridor. The doors of any room can be locked or unlocked only from inside the room (i.e., no door that is locked can be opened from outside the room).

In the figure below, the six rooms and the corridors that connect the six rooms are shown. At the beginning of a particular day, all the doors were unlocked. There were six persons – A through F – each in a different room, as shown in the figure. Any person when going from one room to another or when passing through a room, will lock the door through which they entered the room immediately after they pass through it.

In each room, a screen displays the real-time status of all the doors (to all the rooms) that are locked and unlocked. When travelling from one room to another, any person will use the information about the status of the doors to finalize the route that he will take, ensuring that he locks the minimum possible number of doors in reaching his destination. If there is more than one way in which this can be done, he will choose any one route at random. Also, no person inside a room will unlock any door for any person outside the room.



It is known that on that day A went to R3, after which C went to R5. After this, E went to R4, after which B went to R4 and finally, F went to R4. All of this happened before 12 noon and no one else went to any other room before 12 noon.

Q16. DIRECTIONS *for questions 16 and 17:* Select the correct alternative from the given choices.

At exactly 12 noon, if D wanted to go to R2, how many doors will D lock in going to R2?

☐ a) 1

☐ b) 2

☐ c) 3

☐ d) 4

Q17. DIRECTIONS *for questions 16 and 17:* Select the correct alternative from the given choices.

At exactly 12 noon, if B went to R1 and then, A wanted to go to R4, how many doors will A lock in going to R4?

- ☐ a) 2
- ☐ b) 3
- ☐ c) 1
- ☐ d) It is not possible for A to go to R4.

Q18. DIRECTIONS *for questions 18 and 19:* Type in your answer in the input box provided below the question.

At exactly 12 noon, D went to R1 and then, F went to R6.

Now, if D wanted to go to R3, how many doors will D lock?

Q19. DIRECTIONS *for questions 18 and 19:* Type in your answer in the input box provided below the question.

At exactly 12 noon, D went to R1 and then, F went to R6.

Now, if B wanted to take a chocolate from D and give it to E, and then take a biscuit from E and give it to A, how many doors will B lock in this process?

Q20. DIRECTIONS *for question 20:* Select the correct alternative from the given choices.

At exactly 12 noon, D went to R1 and then, F went to R6.

Now, if C went to R6 and then from R6 to R3, which of the following cannot happen immediately after this?

I. A going to R1

II. F going to R4

III. D going to R3

IV. E going to R2

- ☐ a) Only I and IV
- ☐ b) Only III
- ☐ c) Only I, III and IV
- ☐ d) Only I and II